

EBN 111/122
ELECTRICITY AND ELECTRONICS

Chapter 9
Sinusoidal Steady-State
Analysis

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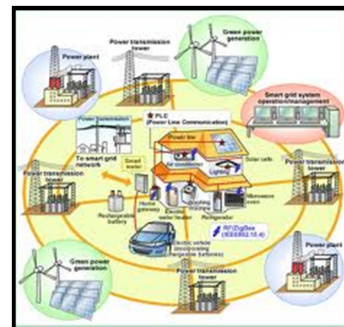
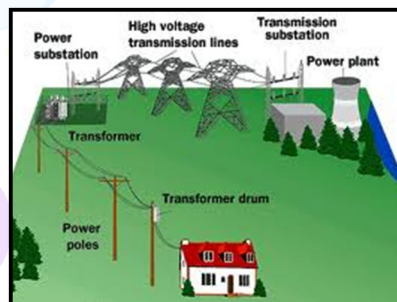
Sinusoids and Phasor
Chapter 9

- 9.1 Motivation
- 9.2 Sinusoids' features
- 9.3 Phasors
- 9.4 Phasor relationships for circuit elements
- 9.5 Impedance and admittance
- 9.6 Kirchhoff's laws in the frequency domain
- 9.7 Impedance combinations

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9.1 Motivation

- AC = Alternating Current (Alternates between + and -)
- Vast majority of Power Networks are AC networks because:
 - AC allows for easy stepping up and down of voltages which allows for efficient transfer of energy over long distances.



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9.1 Motivation

How does one analyse an AC circuit?

Method #1: Time domain

KVL: $-V_s + v_R + v_C + v_L = 0$

$$-10 \cos 4t + iR + \frac{1}{C} \int_{t_0}^t i dt + L \frac{di}{dt} = 0$$

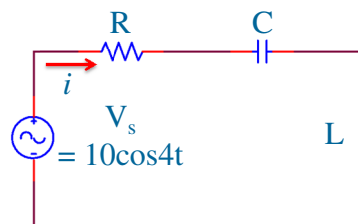
Eliminate $\int$ by $\frac{d}{dt} f(t)$:

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} + 40 \sin 4t = 0$$

2nd order differential equation with solutions:

$$i(t) = A_i \cos(4t + \phi_i) \quad V_C(t) = A_C \cos(4t + \phi_C)$$

$$V_R(t) = A_R \cos(4t + \phi_R) \quad V_L(t) = A_L \cos(4t + \phi_L)$$



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9.1 Motivation

How does one analyse an AC circuit?

Method #2:
Frequency or
Phasor domain

KVL:

$$-\bar{V}_s + \bar{V}_R + \bar{V}_C + \bar{V}_L = 0$$

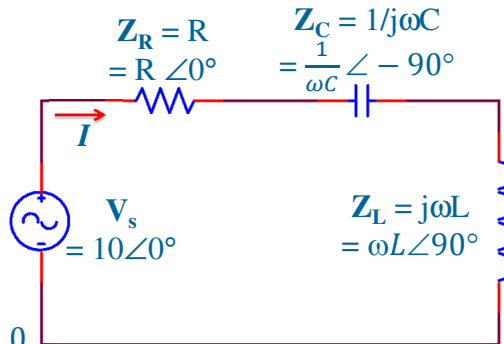
$$-\bar{V}_s + IZ_R + IZ_C + IZ_L = 0$$

$$-10 + I \left(R + \frac{1}{j\omega C} + j\omega L \right) = 0$$

$$I = \frac{10}{R + \frac{1}{j\omega C} + j\omega L} = \frac{10 \angle 0^\circ}{R \angle 0^\circ + \frac{1}{\omega C} \angle -90^\circ + \omega L \angle 90^\circ} = A_i \angle \phi_i^\circ$$

$$A_i \angle \phi_i^\circ \rightarrow i(t) = A_i \cos(\omega t + \phi_i)$$

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9.2 Sinusoids

- A sinusoid is a signal that has the form of the **sine or cosine** function.
- A general expression for the sinusoid:

$$v(t) = V_m \sin(\omega t + \phi)$$

where

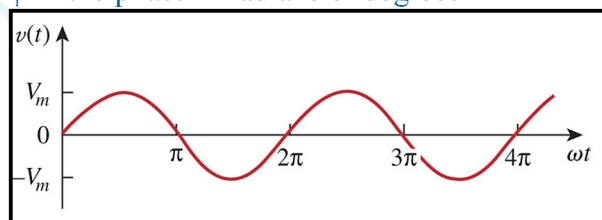
V_m = the **amplitude** of the sinusoid

ω = the angular frequency in radians/s

t = time in seconds

ωt = argument of the sinusoid in radians

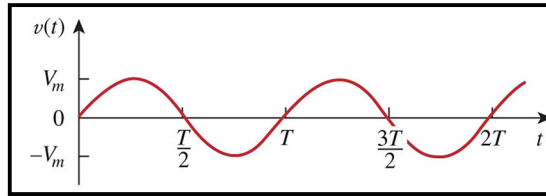
ϕ = the phase in radians or degrees



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9.2 Sinusoids

V vs Time (not ωt)



$T = \text{period [s]}$

At $t = T$: $\omega T = 2\pi$:

$$T = \frac{2\pi}{\omega}$$

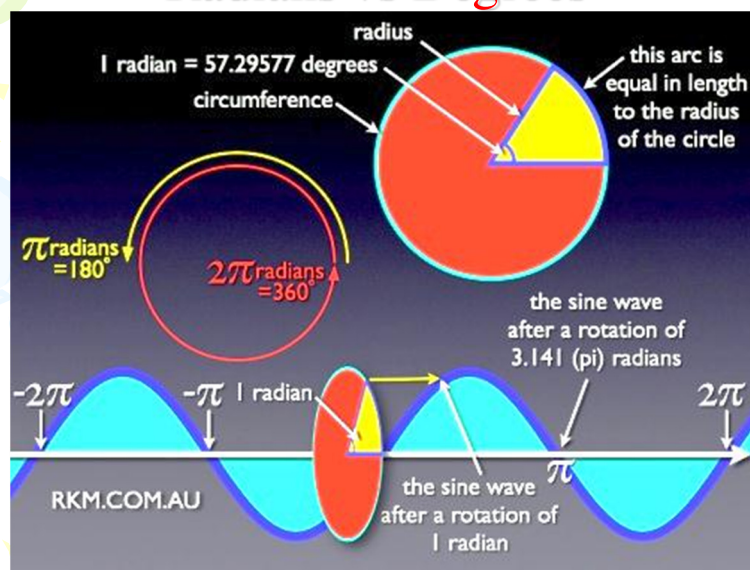
$f = \frac{1}{T}$ frequency in [Hz]

$\omega = 2\pi f$ [rad/s]

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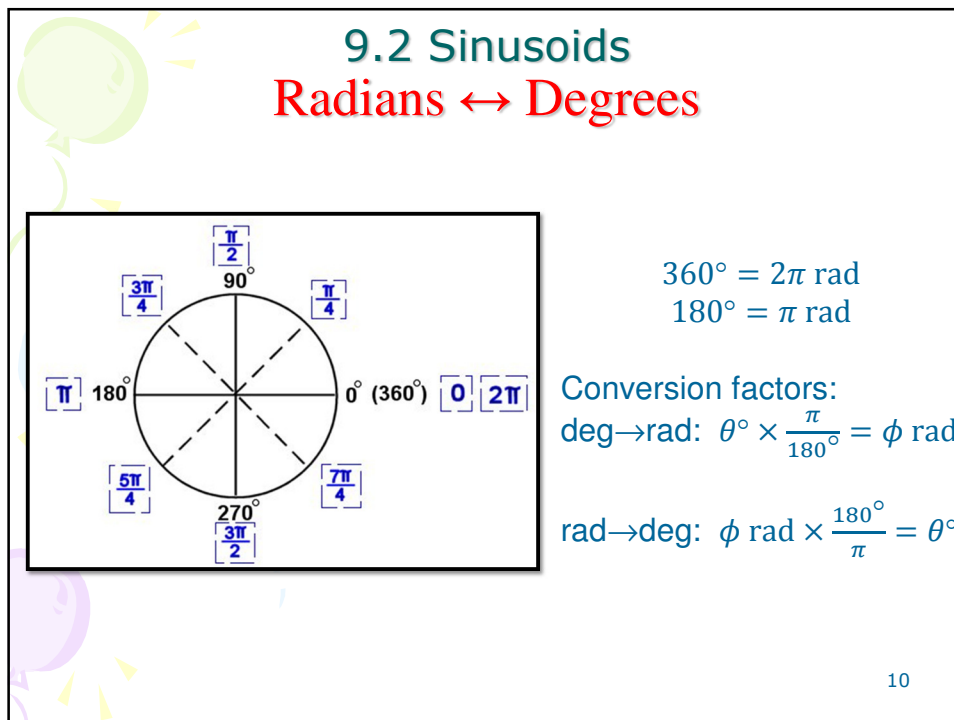
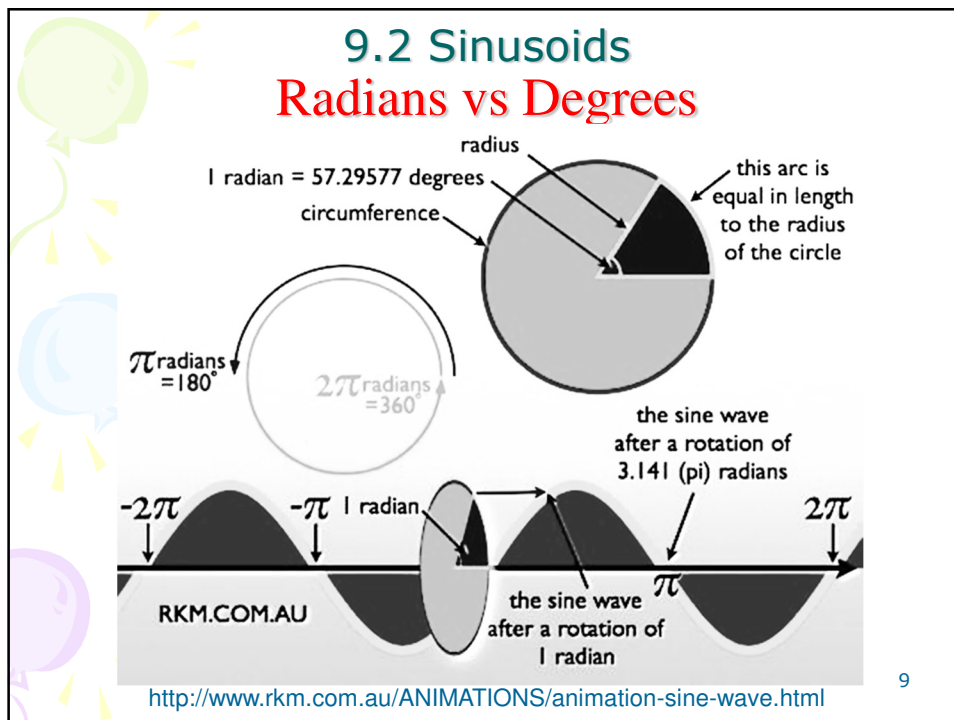
9.2 Sinusoids

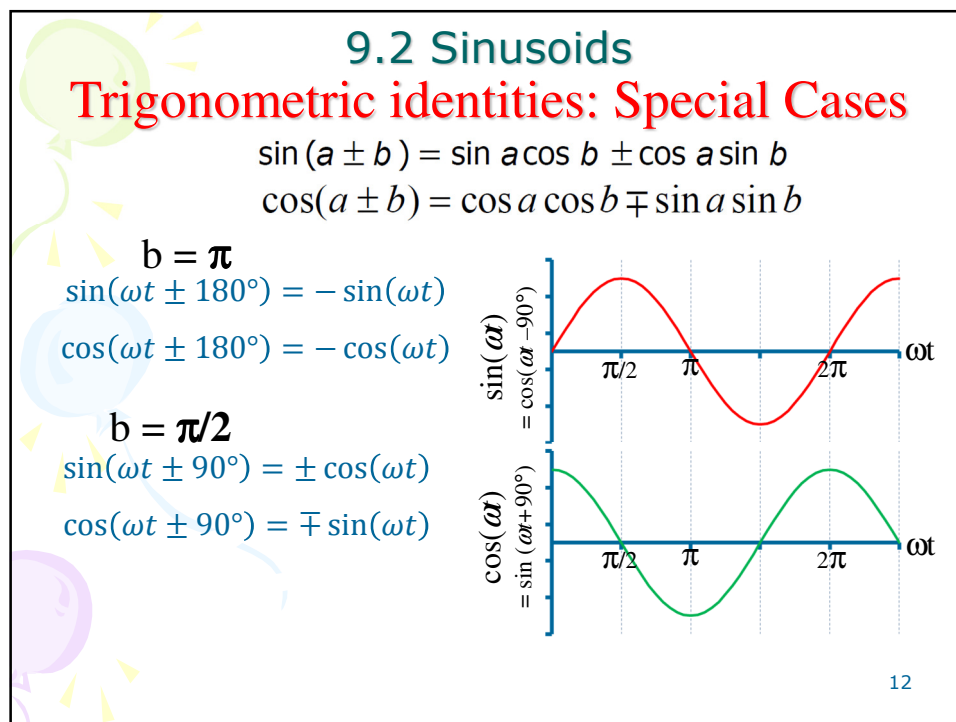
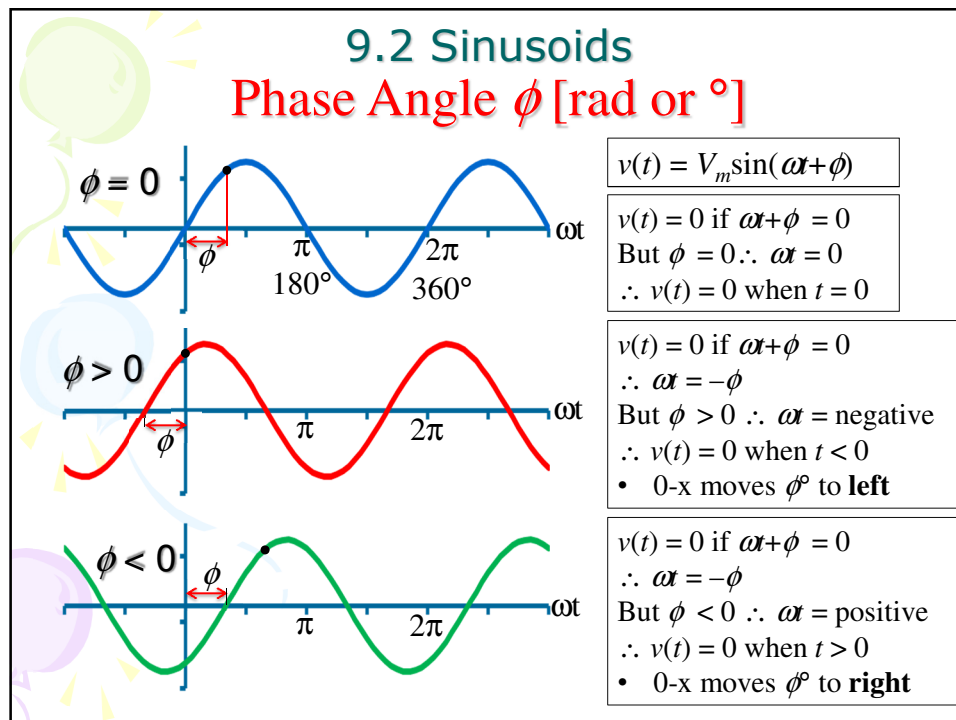
Radians vs Degrees

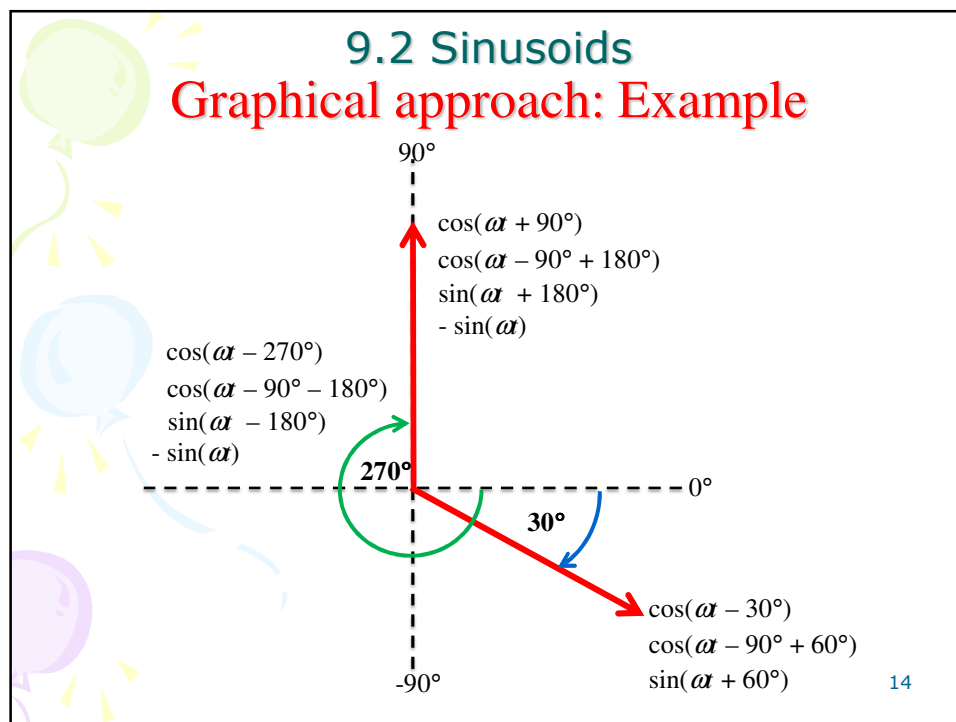
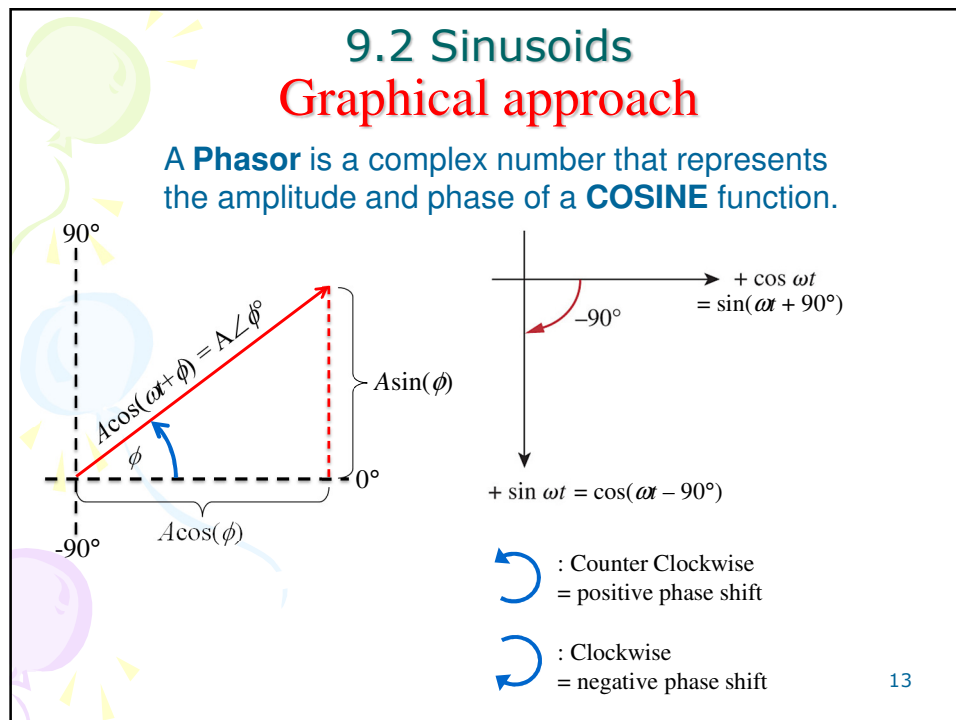


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<http://www.rkm.com.au/ANIMATIONS/animation-sine-wave.html>







PP 9.1

For $f(t) = 5 \sin(4\pi t - 60^\circ)$, determine:

Amplitude; Phase; Angular frequency, Period and Frequency.

Amplitude	
Phase	
Angular frequency	
Frequency	
Period	

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PP 9.2

Determine the phase angle between $i_1 = -4\sin(377t + 25^\circ)$ and $i_2 = 5 \cos(377t - 40^\circ)$.

Analytical:

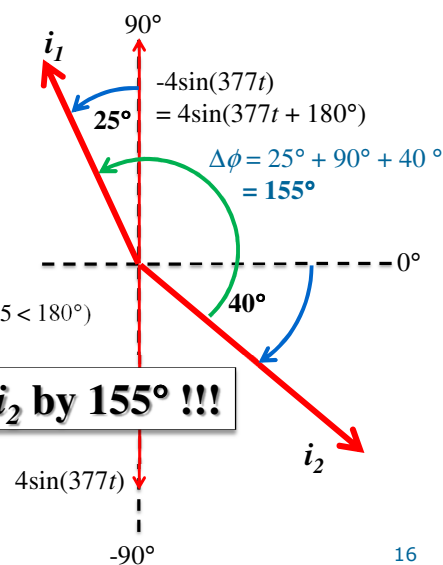
$$\begin{aligned}
 i_1 &= -4\sin(377t + 25^\circ) \\
 &= 4\sin(377t + 25^\circ - 180^\circ) \\
 &= 4\sin(377t - 155^\circ) \\
 &= 4\sin(377t + 90^\circ - 245^\circ) \\
 &= 4\cos(377t - 245^\circ) \\
 &= 4\cos(377t + 115^\circ) \quad (-245 + 360 = 115 < 180^\circ)
 \end{aligned}$$

Hence:

$$\begin{aligned}
 \Delta\phi &= \phi_1 - \phi_2 \\
 &= 115^\circ - (-40^\circ) \\
 &= 155^\circ
 \end{aligned}$$

(Always give angle $< 180^\circ$)

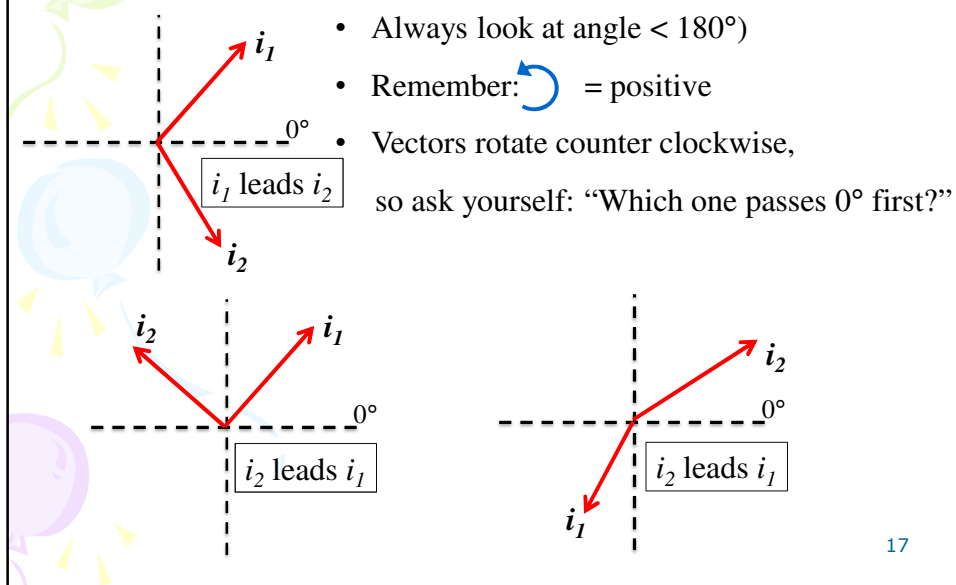
i_1 leads i_2 by 155° !!!



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9.2 Sinusoids

Graphical approach: Which one leads?



Ch. 9 L#2

9.3 Phasors

- A phasor is a complex number that represents the amplitude and phase of a sinusoid (COSINE).

Complex Numbers:

$$\text{Complex Number } \mathbf{z} = \mathbf{x} + j\mathbf{y}$$

Real part
Imaginary part

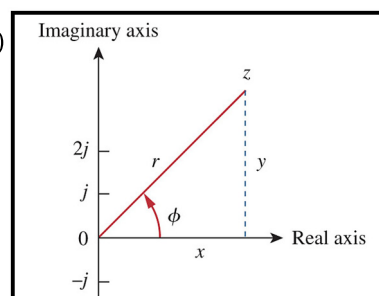
Three different forms of writing:

- Rectangular: $z = x + jy = r(\cos \phi + j \sin \phi)$
- Polar: $z = r \angle \phi^\circ$ (r = magnitude of z)
- Exponential: $z = r e^{j\phi}$

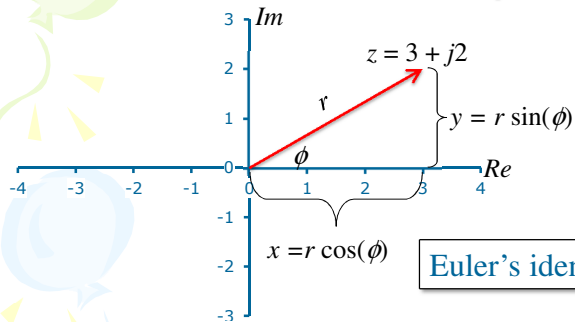
where

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1} \frac{y}{x}$$



9.3 Phasors On the complex plane



$$\begin{aligned}
 z &= x + jy \\
 &= r\cos\phi + j r\sin\phi \\
 &= r(\cos\phi + j\sin\phi) \\
 &= re^{j\phi} \\
 &= r\angle\phi
 \end{aligned}$$

Euler's identity: $e^{\pm j\phi} = \cos\phi \pm j\sin\phi$

A complex number consists out of:

- A Real part (x) and Imaginary part (y):

$$z = x + jy$$

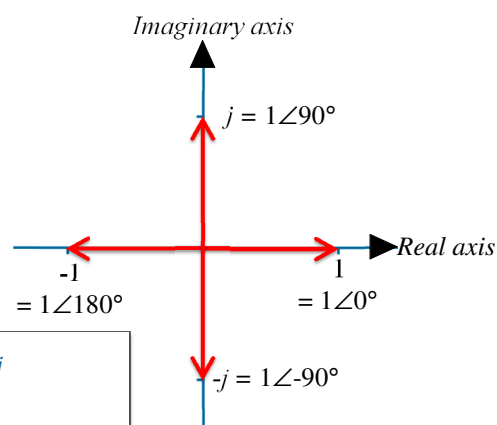
OR

- A Magnitude (r) and a phase (ϕ):

$$z = r\angle\phi = re^{j\phi}$$

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9.3 Phasors On the complex plane



$$\sqrt{-1} = j$$

$$\frac{1}{j} = \frac{1}{1\angle 90^\circ} = 1\angle(-90^\circ) = -j$$

$$1 + j = \sqrt{2}\angle 45^\circ$$

$$1 - j = \sqrt{2}\angle(-45^\circ)$$

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9.3 Phasors Arithmetic (1)

Assume:	$z = x + jy$	$= r \angle \phi^\circ$
	$z_1 = x_1 + jy_1$	$= r_1 \angle \phi_1^\circ$
	$z_2 = x_2 + jy_2$	$= r_2 \angle \phi_2^\circ$

Addition (+): **MUST WORK IN RECTANGULAR FORMAT**

$$\begin{aligned} z_1 + z_2 &= x_1 + jy_1 + x_2 + jy_2 \\ &= (x_1 + x_2) + j(y_1 + y_2) \end{aligned}$$

Subtraction (-): **MUST WORK IN RECTANGULAR FORMAT**

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication (×): **WORK IN POLAR FORMAT**

$$\begin{aligned} z_1 z_2 &= r_1 \angle \phi_1^\circ \cdot r_2 \angle \phi_2^\circ = r_1 r_2 \angle (\phi_1^\circ + \phi_2^\circ) \\ &= r_1 e^{j\phi_1} \cdot r_2 e^{j\phi_2} = r_1 \cdot r_2 e^{j(\phi_1 + \phi_2)} \end{aligned}$$

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9.3 Phasors Arithmetic (2)

Division(÷): **WORK IN POLAR FORMAT**

$$\frac{z_1}{z_2} = \frac{r_1 e^{j\phi_1}}{r_2 e^{j\phi_2}} = \frac{r_1}{r_2} e^{j\phi_1 - \phi_2} = \frac{r_1 \angle \phi_1}{r_2 \angle \phi_2} = \frac{r_1}{r_2} \angle (\phi_1 - \phi_2)$$

Root ($\sqrt{}$): $\sqrt{z} = \sqrt{r} \angle (\phi/2)$

Inverse ($\frac{1}{z}$): $\frac{1}{z} = \frac{1}{r \angle \phi} = \frac{1}{r} \angle (-\phi)$

Conjugate z^* : $\begin{aligned} z^* &= (x + jy)^* = (x - jy) \\ &= r \angle (-\phi) = e^{-j\phi} \end{aligned}$

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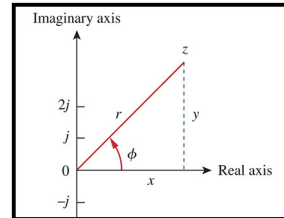
9.3 Phasors

AC signal as a Phasor

Euler's identity: $e^{\pm j\phi} = \cos\phi \pm j\sin\phi$

$$\operatorname{Re}\{e^{j\phi}\} = \cos\phi$$

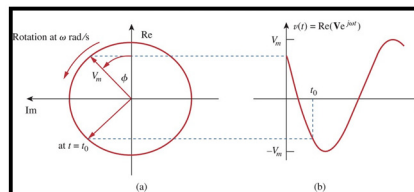
$$\operatorname{Im}\{e^{j\phi}\} = \sin\phi$$



$$\begin{aligned} v(t) &= V_m \cos(\omega t + \phi) = \operatorname{Re}\{V_m e^{j(\omega t + \phi)}\} \\ &= \operatorname{Re}\{V_m e^{j\phi} e^{j\omega t}\} \\ &= \operatorname{Re}\{V e^{j\omega t}\} \end{aligned}$$

$$e^{j\omega t} = \cos\omega t + j\sin\omega t$$

$$\begin{aligned} V &= \bar{V} = V_m e^{j\phi} = V_m \angle \phi \\ &= V_m \cos\phi + jV_m \sin\phi \\ &= x + jy \end{aligned}$$



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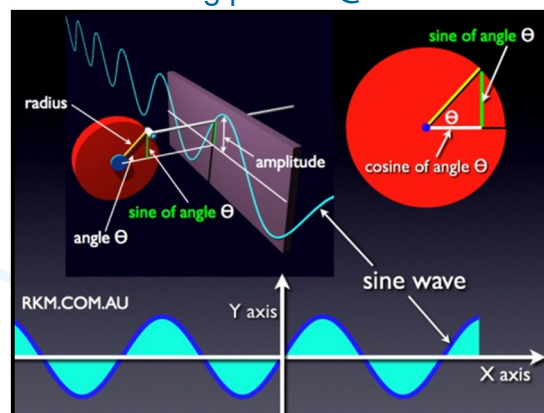
9.3 Phasors

AC signal as a Phasor

$$\operatorname{Re}\{V e^{j\omega t}\}$$

$$e^{j\omega t} = \cos\omega t + j\sin\omega t:$$

A rotating phasor @ ω rad/s



<http://www.rkm.com.au/ANIMATIONS/animation-sine-wave.html>

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9.3 Phasors Transformations

- Transform a sinusoid to and from the time domain to the phasor domain:

$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow V = V_m \angle \phi$$

(time domain)

(phasor domain)

- Amplitude and phase difference are two principal concerns in the study of voltage and current sinusoids.
- Phasor will be defined from the cosine function in all our proceeding study. If a voltage or current expression is in the form of a sine, it will be changed to a cosine by subtracting from the phase.

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9.3 Phasors Sinusoid-phasor Transformation

Time domain representation

Phasor domain representation

$$V_m \cos(\omega t + \phi)$$

$$V_m \angle \phi$$

$$V_m \sin(\omega t + \phi)$$

$$V_m \angle \phi - 90^\circ$$

$$I_m \cos(\omega t + \theta)$$

$$I_m \angle \theta$$

$$I_m \sin(\omega t + \theta)$$

$$I_m \angle \theta - 90^\circ$$

$$\int v dt$$

$$\frac{V}{j\omega}$$

$$\frac{dv}{dt}$$

$$j\omega V$$

$$\begin{aligned} v(t) &= V_m \cos(\omega t + \phi) \\ \frac{dv(t)}{dt} &= \frac{d}{dt} \operatorname{Re} \{ V_m e^{j\phi} e^{j\omega t} \} \\ &= \operatorname{Re} \left\{ V_m e^{j\phi} \frac{d}{dt} e^{j\omega t} \right\} \\ &= \operatorname{Re} \{ V_m e^{j\phi} j\omega e^{j\omega t} \} \end{aligned}$$

$$V j\omega$$

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PP 9.3

Evaluate the following complex numbers:

a) $[(5 + j2)(-1 + j4) - 5\angle 60^\circ]^*$

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PP 9.3

Evaluate the following complex numbers:

b)

$$\frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5$$

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PP 9.4

Express these sinusoids as phasors:

a) $v = 7 \cos(2t + 40^\circ) \text{ V}$

$\mathbf{V} =$

b) $i = -4 \sin(10t + 10^\circ) \text{ A}$

$\mathbf{I} =$

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PP 9.5

Find the sinusoids corresponding to these phasors:

a) $\mathbf{V} = -10 \angle 30^\circ \text{ V}$

$v =$

b) $\mathbf{I} = j(5 - j12) \text{ A}$

$i =$

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Ch. 9 L#3

9.4 Phasors and Circuit Elements

Resistor:

$v = iR$
 $V = IR$

V & I in phase

Inductor:

$v = L \frac{di}{dt}$
 $V = j\omega LI$

**V leads I
I lags V**

Capacitor:

$i = C \frac{dv}{dt}$
 $I = j\omega CV$

**I leads V
V lags I**

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9.4 Phasors and Circuit Elements

Voltage-Current Relationship

Summary of voltage-current relationship		
Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$
L	$v = L \frac{di}{dt}$	$V = j\omega LI$
C	$i = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$

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9.5 Impedance and Admittance

- The impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I , measured in ohms Ω .

$$Z = \frac{V}{I} = R + jX$$

where $R = \text{Re of } Z$ = the resistance and $X = \text{Im of } Z$ = the reactance. **Positive X is for L and negative X is for C .**

- The admittance Y is the reciprocal of impedance, measured in siemens (S).

$$Y = \frac{1}{Z} = \frac{I}{V}$$

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9.5 Impedance and Admittance

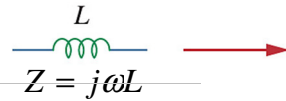
Impedances and admittances of passive elements

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

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9.5 Impedance and Admittance

Remember: $\omega = 2\pi f$



$f = 0 \Rightarrow \text{DC} \Rightarrow Z_L = 0 \Rightarrow \text{Short Circuit}$

$f = \infty \Rightarrow \text{high AC} \Rightarrow Z_L = \infty \Rightarrow \text{Open Circuit}$

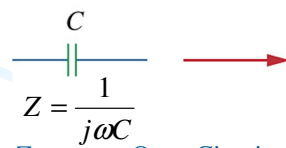
(a)

$\omega = 0; Z = 0$

Short circuit at dc

$\omega \rightarrow \infty; Z \rightarrow \infty$

Open circuit at
high frequencies



$f = 0 \Rightarrow \text{DC} \Rightarrow Z_C = \infty \Rightarrow \text{Open Circuit}$

$f = \infty \Rightarrow \text{high AC} \Rightarrow Z_C = 0 \Rightarrow \text{Short Circuit}$

(b)

$\omega = 0; Z \rightarrow \infty$

Open circuit at dc

$\omega \rightarrow \infty; Z = 0$

Short circuit at
high frequencies

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9.5 Impedance and Admittance



$$Z = R + j\omega L = R + jX_L \quad \therefore X_L = \omega L$$

Impedance

Resistance

Reactance



$$Z = R + \frac{1}{j\omega C} = R - j\frac{1}{\omega C} = R + jX_C \quad \therefore X_C = -\frac{1}{\omega C}$$

$$Z = R + jX = |Z| \angle \theta$$

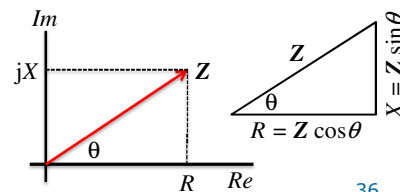
$$|Z| = \sqrt{R^2 + X^2} \quad \theta = \tan^{-1} \frac{X}{R}$$

$$Y = \frac{1}{Z} = G + jB$$

Admittance

Conductance

Susceptance



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9.5 Impedance and Admittance

After we know how to convert RLC components from time to phasor domain, we can transform a time domain circuit into a phasor/frequency domain circuit.

Hence, we can apply the KCL laws and other theorems to directly set up phasor equations involving our target variable(s) for solving.

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PP 9.8

If voltage $v = 10 \cos(100t + 30^\circ)$ V is applied to a $50 \mu\text{F}$ capacitor, calculate the current through the capacitor.

Step 1: Convert from time to phasor domain

$$v = 10 \cos(100t + 30^\circ) \Rightarrow \mathbf{V} = 10 \angle 30^\circ \text{ V}$$

$$\mathbf{Z} = jX_C = j(-1/\omega C) = -j/(100 \times 50 \times 10^{-6})$$

$$= -j200 \Omega$$

$$= 200 \angle -90^\circ \Omega$$

Step 2: Apply usual Laws and Theorems

$$\mathbf{I} = \mathbf{V} / \mathbf{Z} = 10 \angle 30^\circ / 200 \angle -90^\circ$$

$$= 50 \angle (30 - (-90))^\circ \text{ mA}$$

$$= 50 \angle 120^\circ \text{ mA}$$

Step 3: Convert from phasor to time domain

$$i(t) = 50 \cos(100t + 120^\circ) \text{ mA}$$

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PP 9.9Determine $v(t)$ and $i(t)$.

$$v_s = 20 \sin(10t + 30^\circ) \text{ V}$$

Step 1: Convert from time to phasor domain

$$v_s = 20 \sin(10t + 30^\circ) = 20 \cos(10t - 60^\circ)$$

$$\Rightarrow \mathbf{V}_s = 20 \angle -60^\circ \text{ V}$$

$$\mathbf{Z} = R + jX_L = R + j\omega L$$

=

Step 2: Apply usual Laws and Theorems

$$\mathbf{I} = \mathbf{V} / \mathbf{Z} =$$

=

$$\mathbf{V}_L = \mathbf{I} \mathbf{Z}_L =$$

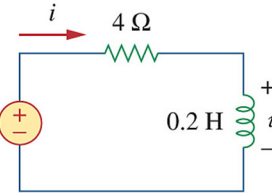
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Step 3: Convert from phasor to time domain

$$i(t) =$$

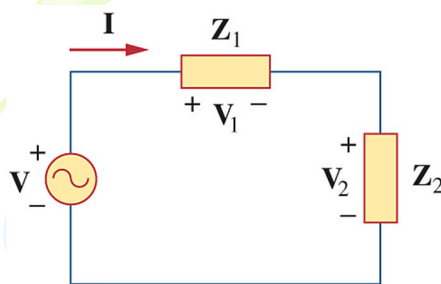
$$v_L(t) =$$

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AC Circuit analysis

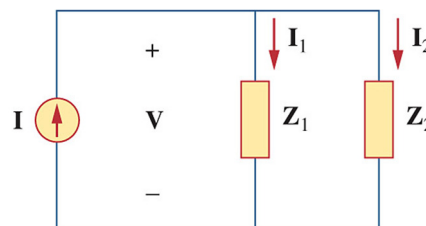
Laws and Theorems



KVL: $-V + V_1 + V_2 = 0$
 $V = IZ_1 + IZ_2$

Z in series: $Z_T = Z_1 + Z_2$

Voltage division: $V_1 = V \frac{Z_1}{Z_1 + Z_2}$

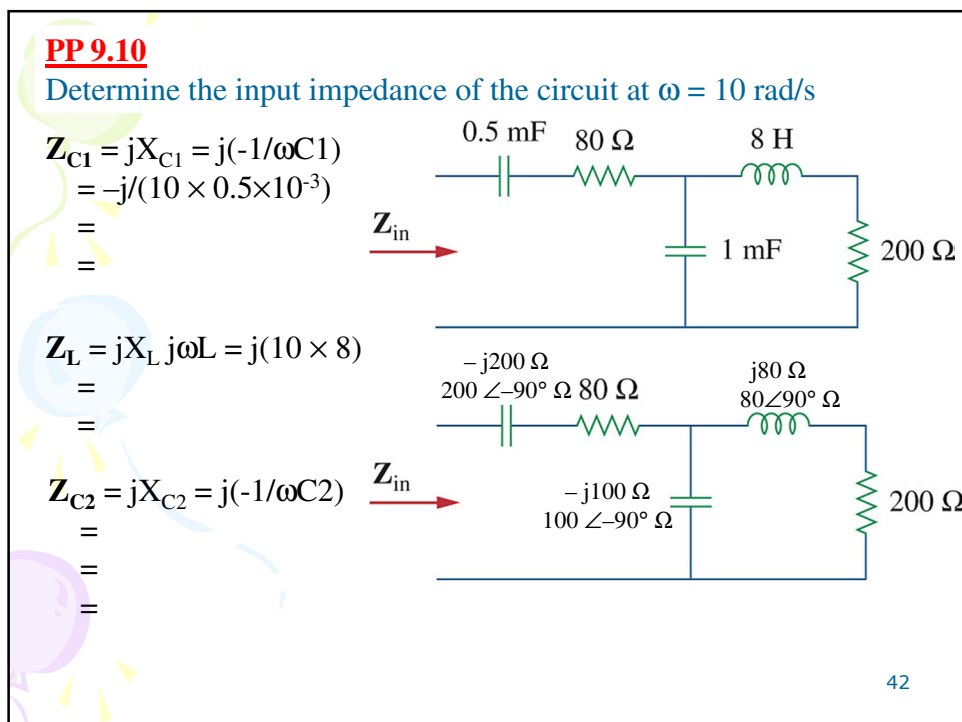
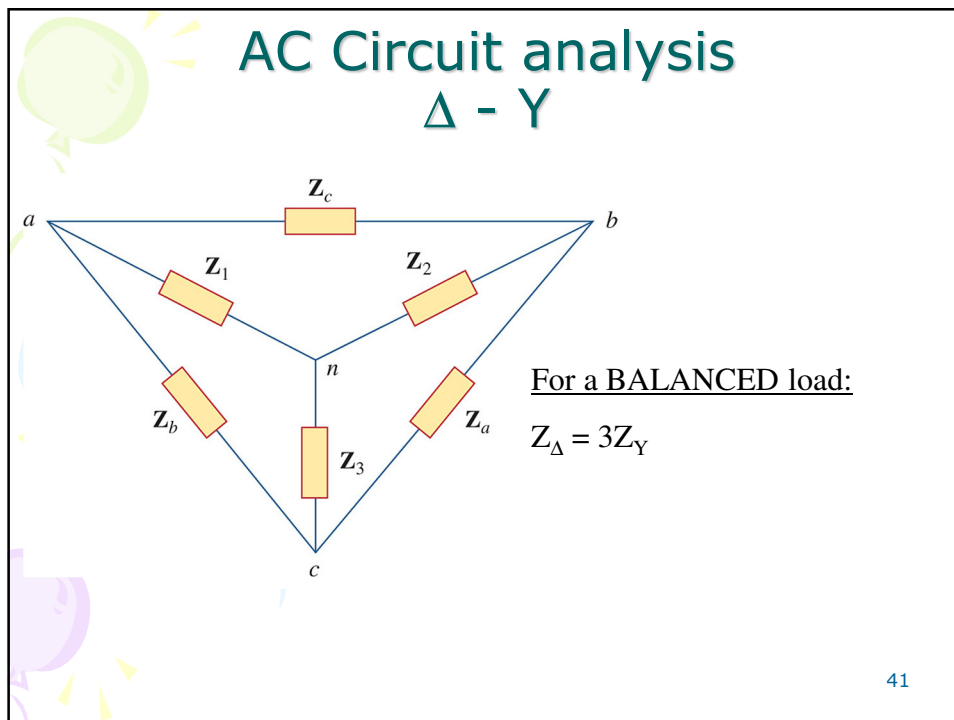


KCL: $I = I_1 + I_2$

Z in //: $Z_{||} = \frac{Z_1 Z_2}{Z_1 + Z_2}$

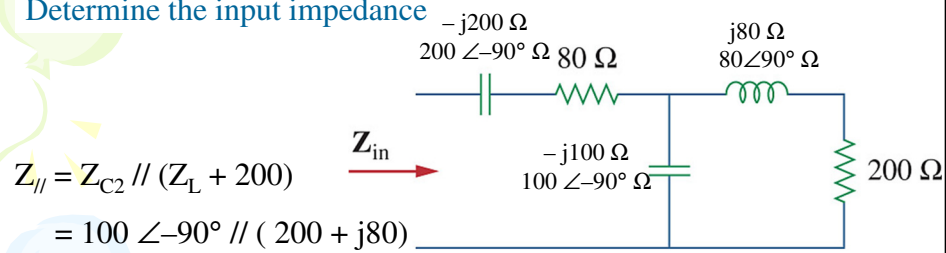
Current division: $I_1 = I \frac{Z_2}{Z_1 + Z_2}$

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PP 9.10 - continued

Determine the input impedance



$$Z_{//} = Z_{C2} // (Z_L + 200)$$

$$= 100 \angle -90^\circ // (200 + j80)$$

$$=$$

$$Z_{in} = Z_{//} + Z_{C1} + 80$$

$$=$$

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